

June 19, 2026

Editor-in-Chief
Symmetry
MDPI

Dear Editor,

I am pleased to submit our manuscript entitled “Symmetry-Aware Attention Networks for Student Engagement Detection: A Lightweight Temporal Approach” for consideration for publication in *Symmetry*.

This paper proposes a symmetry-aware framework for student engagement detection from facial expressions, with three key contributions that connect to the scope of *Symmetry*:

1. **Symmetry-aware attention:** We introduce a multi-scale Channel-Spatial Attention Module (CSAM) that captures both symmetric (engaged) and asymmetric (disengaged) facial patterns through parallel 3×3 , 5×5 , and 7×7 convolutions with adaptive fusion.
2. **Symmetry-breaking detection:** We frame engagement detection as identifying symmetry-breaking phenomena in facial behavior, where engaged states exhibit symmetric muscle activation while disengaged states manifest asymmetric deviations.
3. **Temporal symmetry analysis:** We introduce a BiLSTM-based temporal aggregation module that captures temporal symmetry/asymmetry patterns in video sequences, achieving 94.8% video-level accuracy on DAiSEE.

Key results:

- FER2013: 70.5% accuracy (5-fold CV, 1.26M parameters)
- CK+ cross-dataset: 71.9% accuracy
- DAiSEE: 94.8% video-level accuracy (vs 61.2% frame-level)

This work is directly relevant to *Symmetry* as it addresses symmetry and asymmetry patterns in facial expressions for engagement detection, and proposes symmetry-aware attention mechanisms for efficient edge deployment.

Both authors have read and approved the manuscript. We declare no conflicts of interest.

I look forward to your decision.

Sincerely,

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